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Prop schwartz imp lp

Proposición 1. *La clase de Schwartz* $\mathscr{S}(\mathbb{R})$ *es un espacio vectorial sobre* $\mathbb{C}$ *contenido en* $\mathcal{L}^p(\mathbb{R})$ *para todo* $p \in [1, \infty]$.

***Demostración*:** Que $\mathscr{S}(\mathbb{R})$ es un espacio vectorial es fácil de comprobar. Sea $f \in \mathscr{S}(\mathbb{R})$, entonces $\rho_{0,0} = \sup_{x\in\mathbb{R}} |f(x)| < \infty$, luego $f \in \mathcal{L}^\infty(\mathbb{R})$. Por otro lado, para $p \in [1, \infty)$,

$$\|f\|_p^p = \int_{\mathbb{R}} |f(x)|^p \,\mathrm{d}x = \int_{|x|<1} |f(x)|^p \,\mathrm{d}x + \int_{|x|\geqslant 1} |f(x)|^p \,\mathrm{d}x.$$

Como $f \in \mathcal{L}^\infty(\mathbb{R})$, la primera integral está acotada por $2\,\|f\|_\infty^p$. Para la segunda integral,

$$\int_{|x|\geqslant 1} |f(x)|^p \,\mathrm{d}x = \int_{|x|\geqslant 1} \frac{|x|^{2p}\,|f(x)|^p}{|x|^{2p}} \,\mathrm{d}x \leqslant {\rho_{2,0}}^p \int_{|x|\geqslant 1} \frac{1}{|x|^{2p}} \,\mathrm{d}x = {\rho_{2,0}}^p \cdot 2 \int_1^\infty \frac{1}{x^{2p}} \,\mathrm{d}x < \infty.$$

Por tanto, $\|f\|_p < \infty$ y $f \in \mathcal{L}^p(\mathbb{R})$. ■

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